



Computer Security Principles (Fall 2026) CPSC 4200/6200

Instructor Info —



Mert D. Pesé



Office Hours: Tue 10a-11a



215 McAdams Hall & <https://clemons.zoom.us/my/mpese>



<http://www.mpese.com>



mpese@clemons.edu

Course Info —



Prereq: CPSC 3220 or ECE 3220 and CPSC 3600 or ECE 4380 with a C or better.



Tue & Thu



5.00p-6.15p



G66 Humanities Hall

TA Info —



Yu Wei (Sam) Liu



Office Hours: Wed 10a-11a & Thu 10a-11a



<https://clemons.zoom.us/my/yuweiliu>



yuwei@clemons.edu



Alkim Domeke



Office Hours: Tue 2p-3p



<https://clemons.zoom.us/j/94078099119?pwd=rd2bqfoa3a0yS9btCYqWbRHgkIN7Sy.1>



adomeke@clemons.edu

Course Description

This course teaches the security mindset and introduces the principles and practices of computer security as applied to software, host systems, and networks. It covers the foundations of building, using, and managing secure systems. The four major topics of this course consist of (i) an introduction to cryptography, (ii) web and network security, (iii) host and application security, as well as (iv) examples of real-world security. Lectures include fundamental concepts and principles of software security, operating system and network security, firewalls and intrusion detection systems, private and public key cryptographic algorithms, hash functions, authentication, TLS and web security.

The knowledge obtained from lectures will be deepened by working on homework projects. Students will implement and learn about attacks and defenses that have been either discussed in the lecture or are an extension of the lecture content. As a result, students will obtain a better understanding of fundamental security concepts. Besides the homework projects, there will be three exams (two midterms and one final) and online quizzes at the beginning of each lecture to check understanding.

(Tentative) Grading Scheme

50%

Homework Projects: There will be four projects, which will count for a total of 60% of your course grade. Project teams can consist of up to two students. Working alone is permitted, but highly discouraged due to the workload. You can only team up with students from your course (e.g., a student taking CPSC 4200 can only work with another student taking CPSC 4200). Students taking CPSC 6200 will have additional assignments in each homework to work on. You may consult general reference material, but the material you turn in must be entirely your own work, and you are bound by the Academic Integrity policies of Clemson University.

45%

Exams: There will be three in-person, closed-book exams which are listed in the course schedule below. Each midterm counts 10% of the overall grade and the final 15%. The first exam is a midterm that covers Lectures 1–7, the second exam is a midterm that covers Lectures 8–15 and finally, the third exam is a final that is cumulative, i.e., covers Lectures 1–24.

5%

Participation: At the beginning of each lecture (22 in total), there will be a short quiz on Canvas. Students will get 5 minutes at the beginning of each lecture to submit the quiz online. Students will be allowed to drop/skip 6 quizzes.

Required Materials

The official textbook for the class is "Security Engineering: A Guide to Building Dependable Distributed Systems 3rd Edition" by Ross Anderson (ISBN 978-1119642787). This textbook is not required, but recommended. Slides will be provided on Canvas on the same day of the class.

Learning Objectives

- Explain basic principles and practices of computer security
- Develop a conceptual vocabulary for applied cryptography
- Explain common software and network vulnerabilities and attacks, as well as defense mechanisms against network attacks, and cryptographic protection mechanisms
- Describe the cost and tradeoffs associated with designing security into a product
- Implement and learn about existing attacks by working with a partner
- Learn about real-world security in emerging systems

Grading Policies

All assignments should be submitted through the course website (Canvas), unless stated otherwise. Homework and Projects are due at class time on the due date described in the assignments. Late submissions are highly discouraged. Late policy is as follows: An immediate deduction of 10% will be applied to submissions the second after the deadline. Furthermore, 15% will be deducted every 4 hours, but before 24 hours past the deadline; submissions made 24 hours or more past the deadline will not be accepted. If you feel that an error has been made in grading an assignment or an exam, you must submit a written re-grade petition within one week after the assignment or exam has been returned to you. Verbal appeals are not acceptable and grades will not be changed after the one-week period.

Letter grades are assigned according to the standard 10-point scale:

A	90-100
B	80-89
C	70-79
D*	60-69
F	0-59

The final score will be rounded up, e.g., if you receive a 89.5, it will be rounded up to 90 and you will still receive an A. There might be extra credit opportunities throughout the semester which will be announced on Canvas if necessary. Curving is at the discretion of the professor. Letter grade conversion for CPSC 6200 might differ from CPSC 4200.

* For CPSC 4200 only.

Time to Wait

Students are expected to wait for 15 minutes after the beginning of a lecture before leaving if the instructor is late and no announcement has been posted on Canvas.

Absences

As per Clemson academic policy, course instructors may use reasonable academic penalties which reflect the importance of work missed due to unexcused absences, since absence from class is detrimental to the learning process. Course instructors who penalize students for unexcused absences must state attendance requirements as related to the grading, on the course syllabus and keep accurate attendance records. See the Academic Regulations section of the current Undergraduate catalog or the Policies and Procedures of the Graduate catalog, both located at the Course Catalog home page.

If you are sick, please stay home. Let the professor know in advance via e-mail that you are not attending that day's class and you are fine. If there are other reasons (e.g., family-related, etc.) that you cannot attend, please let the professor know as well.

Academic Integrity

As members of the Clemson University community, we have inherited Thomas Green Clemson's vision of this institution as a "high seminary of learning." Fundamental to this vision is a mutual commitment to truthfulness, honor, and responsibility, without which we cannot earn the trust and respect of others. Furthermore, we recognize that academic dishonesty detracts from the value of a Clemson degree. Therefore, we shall not tolerate lying, cheating, or stealing in any form.

See the Undergraduate Academic Integrity Policy website for additional information and the current catalogue for the policy. For graduate students, see the current Graduate School Handbook for all policies and procedures.

Accessibility

Clemson University values the diversity of our student body as a strength and a critical component of our dynamic community. Students with disabilities or temporary injuries/conditions may require accommodations due to barriers in the structure of facilities, course design, technology used for curricular purposes, or other campus resources. Students who experience a barrier to full access to this class should let the instructor know and make an appointment to meet with a staff member in Student Accessibility Services as soon as possible. You can make an appointment by calling 864-656-6848, by emailing studentaccess@lists.clemson.edu, or by visiting Suite 239 in the Academic Success Center building. Appointments are strongly encouraged – drop-ins will be seen, if at all possible, but there could be a significant wait due to scheduled appointments. Students who have accommodations are strongly encouraged to request, obtain and send these to their instructors through the AIM portal as early in the semester as possible so that accommodations can be made in a timely manner. It is the student's responsibility to follow this process each semester.

You can access further information at the Student Accessibility website. Other information is at the university's Accessibility Portal.

The Clemson University Title IX Statement Regarding Non-Discrimination

The Clemson University Title IX statement: Clemson University is committed to a policy of equal opportunity for all persons and does not discriminate on the basis of race, color, religion, sex, sexual orientation, gender, pregnancy, national origin, age, disability, veteran's status, genetic information or protected activity in employment, educational programs and activities, admissions and financial aid. This includes a prohibition against sexual harassment and sexual violence as mandated by Title IX of the Education Amendments of 1972. This Title IX policy is located on the Campus Life website. Ms. Alesia Smith is the Clemson University Title IX

Clemson University aspires to create a diverse community that welcomes people of different races, cultures, ages, genders, sexual orientation, religions, socioeconomic levels, political perspectives, abilities, opinions, values and experiences.

Emergency Preparation

Emergency procedures have been posted in all buildings and on all elevators. Students should be reminded to review these procedures for their own safety. All students and employees should be familiar with guidelines from the Clemson University Police Department. Visit [here](#) for information about safety.

Clemson University is committed to providing a safe campus environment for students, faculty, staff, and visitors. As members of the community, we encourage you to take the following actions to be better prepared in case of an emergency:

1. Ensure you are signed up for emergency alerts
2. Download the Rave Guardian app to your phone
3. Learn what you can do to prepare yourself in the event of an active threat

Your Well-being is Important

As a student you may experience a range of personal issues that can impede learning, such as strained relationships, increased anxiety, alcohol/drug concerns, feeling down, sadness, difficulty concentrating, lack of motivation, or other issues. These mental health concerns may impact your academic performance or your participation in daily activities. It is very important that you ask for help when you are struggling. Please reach out to me or to Clemson's mental health services to guide you to resources that will help.

Changes to Syllabus

Lecture topics and assignments are subject to change. The course syllabus is a general plan for the course; deviations to the class may be necessary and will be announced to class by the instructor.

Ethics and Law

To defend a system, you need to be able to think like an attacker, and that includes understanding techniques that can be used to compromise security. However, using those techniques in the real world may violate the law or the university's rules, and it may be unethical. Under some circumstances, even probing for weaknesses may result in severe penalties, up to and including expulsion, civil fines, and jail time. Our policy in this course is that you must respect the privacy and property rights of others at all times, or else you will fail the course.

Acting lawfully and ethically is your responsibility. Carefully read the Computer Fraud and Abuse Act (CFAA), a federal statute that broadly criminalizes computer intrusion. This is one of several laws that govern "hacking." Understand what the law prohibits. The EFF provides helpful advice on vulnerability reporting and other legal matters.

AI Statement

The use of artificial intelligence (AI) tools are not permitted for any work in this course, even with proper documentation and citation.

Course Schedule

Date	Topic	Other Notes
08/20 (R)	Course Introduction & Lecture 1: Security Mindset	N/A
08/25 (T)	Lecture 2: Message Integrity (Part I)	N/A
08/27 (R)	Lecture 3: Message Integrity (Part II)	HW1 released
09/01 (T)	Lecture 4: Confidentiality (Part I)	N/A
09/03 (R)	Lecture 5: Confidentiality (Part II)	N/A
09/08 (T)	Lecture 6: Key Exchange	N/A
09/10 (R)	Lecture 7: Public Key Cryptography	HW1 due
09/15 (T)	Midterm I Recap: Cryptography	HW2 released
09/17 (R)	Midterm Exam I: Cryptography	Covers Lectures 1–7
09/22 (T)	Lecture 8: Web Security - Introduction (Part I)	Midterm I Debrief
09/24 (R)	Lecture 9: Web Security - Introduction (Part II)	N/A
09/29 (T)	Lecture 10: Web Security - Attacks	HW2 due
10/01 (R)	Lecture 11: HTTPS (Part I)	HW3 released
10/06 (T)	Lecture 12: HTTPS (Part II)	N/A
10/08 (R)	Lecture 13: Networking 101	N/A
10/13 (T)	Fall Break (No Class)	N/A
10/15 (R)	Lecture 14: Network Attacks	N/A
10/20 (T)	Lecture 15: Network Defenses	N/A
10/22 (R)	Midterm II Recap: Network Security	HW3 due
10/27 (T)	Midterm II Exam: Network Security	Covers Lectures 8–15
10/29 (R)	Lecture 16: Control Hijacking (Part I)	Midterm II Debrief
11/03 (T)	Election Day (No Class)	N/A
11/05 (R)	Lecture 17: Control Hijacking (Part II)	N/A
11/10 (T)	Lecture 18: Control Hijacking (Part III)	HW4 released
11/12 (R)	Lecture 19: Authentication and Passwords	N/A
11/17 (T)	Lecture 20: Malware	N/A
11/19 (R)	Lecture 21: Access Control and Isolation	N/A
11/24 (T)	Lecture 22: Digital Forensics & Side Channels	N/A

11/26 (R)	Thanksgiving Break (No Class)	N/A
12/01 (T)	Lecture 23: Privacy (Part I)	N/A
12/03 (R)	Lecture 24: Privacy (Part II) & Final Recap	HW4 due
12/10 (R)	Final Exam	7:00–9:30p.m.; Covers Lectures 1–24